

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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Anonymous submission

ABSTRACT

Vision-language models (VLMs) have achieved substantial progress across a broad spectrum of visual question answering benchmarks, including visual reasoning, document understanding, and multimodal dialogue. These improvements are visible across VLM families that differ in their base models, alignment architectures, and training corpora. Nevertheless, recent evidence indicates that such models continue to lag on conventional image-classification benchmarks, which more directly probe fine-grained visual knowledge. We therefore evaluate a large collection of recent VLMs on fine-grained classification benchmarks and analyze potential sources of the discrepancy between fine-grained knowledge and performance on other vision benchmarks. Through a sequence of ablation experiments, we find that upgrading the LLM improves all benchmark scores in a broadly uniform manner, whereas strengthening the vision encoder disproportionately benefits fine-grained classification. We further show that the pretraining stage is indispensable for fine-grained performance, especially when the language-model weights are unfrozen during pretraining. Collectively, these results delineate practical directions for improving fine-grained visual understanding and other vision-centric capabilities in VLMs.

1 INTRODUCTION

Recent advances in vision-language models (VLMs), which couple vision encoders with large language models (LLMs), have yielded systems with increasingly sophisticated capacities for interpreting, contextualizing, and reasoning over visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Such models now address complex tasks, including open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023); nevertheless, their apparent breadth leaves unresolved a more elementary but consequential question: how reliably do VLMs perform fine-grained visual perception, and through which modeling or training choices can that capability be improved?

Fine-grained visual recognition—the capacity to discriminate among visually proximate categories—requires models to attend to subtle diagnostic cues while remaining robust to intra-class variation in pose, illumination, and background. Although traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have demonstrated strong performance on fine-grained classification benchmarks, the corresponding behavior of newer VLMs has been comparatively underexamined.

Characterizing fine-grained knowledge in VLMs is essential because perceptual fidelity constitutes a prerequisite for more elaborate forms of understanding and reasoning. Real-world deployments frequently involve long-tailed, fine-grained distributions in which the distinction between closely related categories is not merely semantic but operationally decisive: if a model misidentifies a mushroom species, for example, its textual knowledge cannot reliably support subsequent judgments about toxicity or safe consumption. Moreover, identifying the factors that mediate fine-grained knowledge acquisition can inform the design of more effective architectures and training strategies for future VLMs.

In this paper, we first provide a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs by comparing performance on fine-grained classification benchmarks with capa-

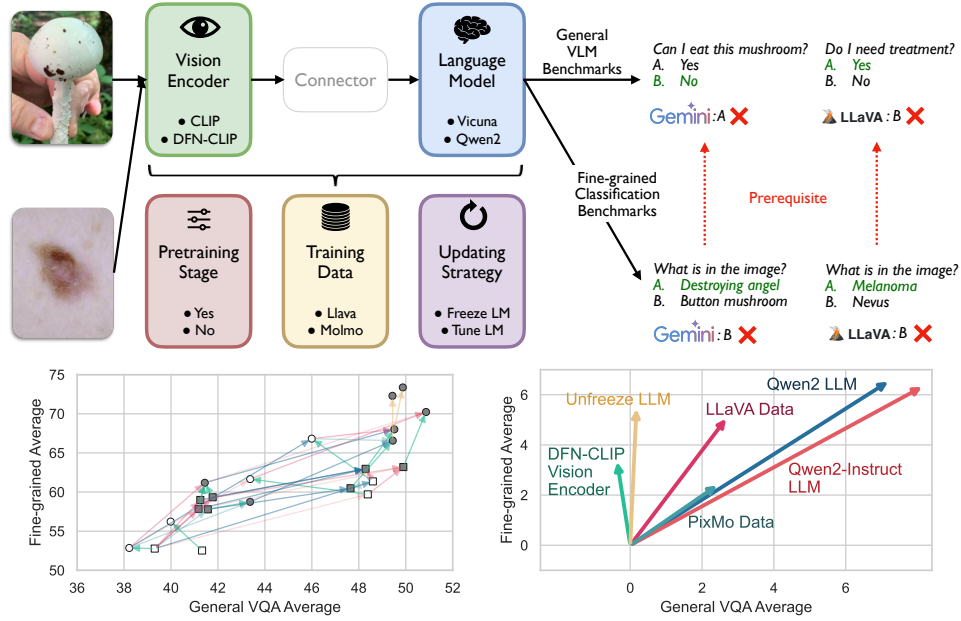


Figure 1: Overview. (Top) We examine the fine-grained classification capacities of vision-language models (VLMs), an essential but frequently underexamined competence that supports higher-level understanding and reasoning. (Bottom) By conducting 22 systematic ablations over central model components and training strategies (left), we isolate the factors that most strongly govern fine-grained classification performance in VLMs (right).

bility estimates from general VLM tasks. Concretely, we evaluate 15 VLMs, including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), Molmo (Deitke et al., 2024), on four fine-grained classification benchmarks—ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014)—alongside aggregated results from eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). The resulting analysis reveals substantial variance in fine-grained classification among models with comparable general abilities, suggesting that fine-grained recognition constitutes a distinct axis of visual intelligence that current VLM benchmarks do not adequately capture. We further observe a persistent gap between VLMs and their CLIP-based vision encoders, echoing prior findings (Zhang et al., 2024; Geigle et al., 2024).

To determine which factors contribute to fine-grained classification performance in VLMs, we conduct a systematic study of major model components and training paradigms. Specifically, we ask whether architectural components, including the language model and vision encoder, differentially affect classification performance, and whether training choices, including pretraining and variation in data quality, modulate the resulting behavior. Across 22 carefully controlled ablation experiments, we identify several factors that materially shape fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Strengthening the base language model yields broadly uniform improvements across fine-grained classification and general VLM benchmarks.
- **Vision encoder (§4.2):** A stronger vision encoder produces pronounced gains in fine-grained classification while exerting a more limited influence on general VLM benchmarks, particularly for models trained with a two-stage recipe.
- **Pretraining stage (§4.3):** Large-scale pretraining on image-captioning datasets substantially improves fine-grained classification, although its effect on general VLM benchmarks is comparatively attenuated.
- **Weight updating method (§4.3):** Fine-grained classification improves when pretraining updates both the connector between the vision encoder and LLM and the LLM itself, rather than restricting optimization to the connector alone.
- **Data quality (§4.3):** The measured influence of pretraining and instruction-tuning data quality on fine-grained classification is limited, especially when language-model weights remain frozen during pretraining.

In summary, our study foregrounds fine-grained visual perception in VLMs as an essential yet insufficiently measured component of current benchmark evaluations. By pairing broad model evaluation with systematic ablation, we identify the architectural and training factors that govern fine-grained knowledge acquisition. These findings provide guidance for developing more vision-centric VLMs with stronger visual understanding, and they transfer naturally to real-world long-tail applications in which fine-grained recognition is indispensable.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) constitute a broad class of systems that integrate visual inputs with language models, often instantiating the LLaVA-style architecture (Liu et al., 2023) composed of three central components: a vision encoder, a language model, and an MLP connector that mediates between them. These models have attained strong performance across heterogeneous multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Correspondingly, a large benchmark ecosystem has emerged, including MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). Yet these benchmarks predominantly assess reasoning and language understanding conditioned on visual inputs, while leaving more vision-centric competencies, such as object recognition and fine-grained classification, comparatively undermeasured. To address this gap, we systematically evaluate and analyze VLMs on fine-grained classification, an essential but still underexplored dimension of visual intelligence.

Fine-grained visual classification. Fine-grained classification is a long-standing computer-vision problem concerned with distinguishing visually similar subcategories embedded within broader object classes, including bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). Benchmarks of this kind have been central to the development and assessment of vision encoders such as ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Despite their importance in vision research, these datasets are largely absent from recent VLM evaluation protocols, which instead prioritize visual question answering. Because many real-world applications involve long-tailed distributions and require precise recognition—for example in medical diagnosis, food safety, and species identification—evaluating VLMs on fine-grained benchmarks is necessary for understanding their practical utility from a vision-centric perspective that extends beyond language-centric reasoning.

Evaluation and ablation of VLMs for fine-grained classification. Although recent studies have begun to evaluate VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), their scope has typically been constrained to a small number of models or datasets. We extend these efforts by evaluating a broader set of VLMs across multiple fine-grained classification benchmarks under a unified multiple-choice formulation. In addition, we move beyond measurement alone by conducting a comprehensive ablation analysis designed to isolate the factors that influence fine-grained classification performance.

Several recent works have examined the broader VLM design space, including model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). However, these analyses are primarily anchored in general benchmarks that emphasize language-mediated reasoning. Here, we investigate how key design choices—including base-model selection, pretraining methodology, and the composition of pretraining and instruction-tuning data—affect fine-grained classification, thereby providing evidence about how VLMs can be optimized for vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section motivates the evaluation of fine-grained classification, specifies the benchmark conversion and evaluation protocol, and presents the principal observations obtained from testing 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a foundational computer-vision task that requires discrimination among visually similar subcategories within a broader category, such as distinct bird species or car models (Wei et al., 2021). Numerous benchmarks, including ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008), have therefore become standard testbeds for training and evaluating vision encoders. With the emergence of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have reached near- or superhuman performance on challenging fine-grained classification tasks.

Nevertheless, these benchmarks have been largely overlooked in the evaluation of vision-language models (VLMs). Although recent VLMs such as LLaVA (Liu et al., 2023) demonstrate strong capacities for interpreting visual concepts, their assessment is usually concentrated on visual question answering (VQA), including academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document-understanding tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These benchmarks foreground reasoning and language processing but contain relatively few fine-grained object categories, leaving core vision-centric competencies only partially examined.

Fine-grained classification is particularly consequential for VLMs because real-world applications frequently encounter long-tailed distributions in which similar categories must be separated through precise visual knowledge. For example, if a blind user asks whether a mushroom is safe to eat, the model must first identify the species correctly before reasoning about toxicity; misclassifying a deadly destroying angel as an edible white button mushroom could have fatal consequences. Analogously, in medical diagnosis, failure to distinguish between visually similar diseases may produce inappropriate treatment recommendations and serious health risks. In autonomous driving, confusing a “stop” sign with a “do not enter” sign—both red and geometrically similar—could induce dangerous navigation errors. Consequently, robust fine-grained classification is a prerequisite for safety and reliability in many real-world VLM scenarios.

3.2 BENCHMARKS

We evaluate the fine-grained visual knowledge of VLMs using four well-established object-recognition benchmarks, each corresponding to a distinct domain of fine-grained classification: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

Because these benchmarks contain between 37 and 1,000 distinct classes, direct evaluation in the multiple-choice format commonly used for VLM training and assessment is nontrivial. To accommodate this constraint, we follow the methodology of Geigle et al. (2024) and convert each dataset into 5-way multiple-choice questions. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest negative labels according to image-label cosine similarity, and these labels are combined with the ground-truth label to form the answer set. We use an OpenCLIP model to avoid biasing the difficulty against VLMs based on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), all of which are trained on datasets different from the OpenCLIP training data. This multiple-choice conversion permits evaluation in a format native to VLMs while retaining much of the original classification difficulty. For instance, CLIP ViT-L/14-336px (Radford et al., 2021) obtains 75.2% accuracy on the 102-class Flowers-102 benchmark, which increases only to 78.3% on our 5-way multiple-choice version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use the corresponding default prompt format from VLMEvalKit (Duan et al., 2024) and compute accuracy from exact matches to the multiple-choice options. For baseline comparison, we evaluate several CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) under the standard zero-shot protocol, in which images and class-label prompts (“A photo of a

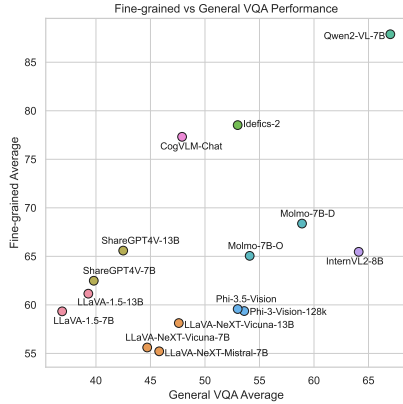


Figure 2: **Fine-grained classification relative to general VQA performance across VLMs.** The distribution of recent VLMs indicates that fine-grained classification constitutes a distinct visual capability that standard VQA benchmarks do not adequately measure.

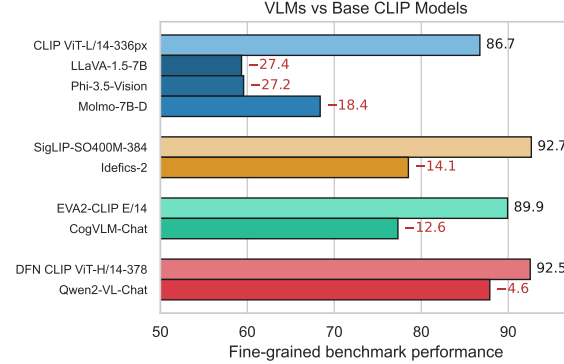


Figure 3: **Comparison between VLMs and their corresponding CLIP vision encoders on fine-grained classification.** Although Qwen2-VL-Chat nearly matches the performance of its DFN-CLIP vision encoder, the remaining VLMs trail their encoders by substantial margins, indicating considerable headroom for improving fine-grained classification.

[class]”) are encoded with CLIP’s image and text encoders. The top-1 prediction is then determined by the cosine similarity between image and text embeddings.

To assess whether existing VLM benchmarks capture fine-grained classification capability, we compare VLM performance on our fine-grained classification benchmarks with each model’s average score on eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

3.4 RESULTS

Figure 2 depicts the relationship between fine-grained classification performance, averaged over our four benchmarks, and general VQA performance, averaged over eight existing VQA benchmarks, for all evaluated VLMs. Models with similar general VQA performance nevertheless exhibit substantial variation in fine-grained classification accuracy. For example, although CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both obtain nearly 48% average accuracy on general VQA benchmarks, their fine-grained classification accuracies differ by 19pp (percentage points)—77.3% for CogVLM-Chat versus 58.1% for LLaVA-NeXT-Vicuna-13B. This discrepancy indicates that existing VLM evaluation benchmarks do not adequately capture fine-grained classification capability, thereby positioning fine-grained classification as a distinct dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Because current VLMs are each built on an existing pretrained vision encoder, our multiple-choice formulation enables a direct comparison between VLMs and CLIP models on traditional classification benchmarks. Figure 3 compares selected VLMs with their corresponding CLIP baselines and reveals a substantial performance gap. For instance, Molmo (Deitke et al., 2024), a recent model that performs strongly on several VQA benchmarks, reaches only 68.4% average accuracy in our multiple-choice evaluation setting, whereas its CLIP ViT-L/14 (Radford et al., 2021) vision encoder obtains 86.7%. Even the best evaluated model, Qwen2-VL-Chat (Wang et al., 2024b), remains 4.6 points behind its own state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder. This gap motivates a closer analysis of VLM design choices, with the goal of identifying the factors that enhance fine-grained classification; accordingly, the next section presents an ablation study of these factors.

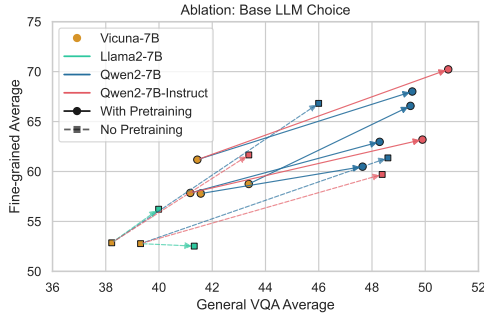


Figure 4: **Ablating the base LLM by replacing Vicuna-7B in LLaVA with alternative LLMs.** On average, substituting Qwen2-7B for Vicuna produces a +7.5pp increase in fine-grained performance and a +7.7pp improvement in general VQA performance.

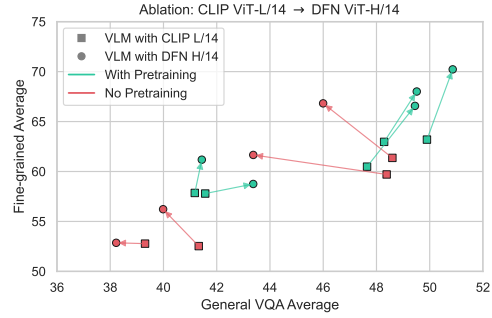


Figure 5: **Ablating the base vision encoder by replacing LLaVA’s CLIP ViT-L/14 with DFN-CLIP ViT-H/14.** With sufficient pretraining, the DFN-CLIP substitution improves fine-grained performance by +4.5pp and general performance by +1.2pp.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational results in Section 3, we ablate salient inter-model differences that may contribute to fine-grained classification performance, with primary emphasis on base-model selection and training methodology. All experiments use the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023) as the common substrate. Each model undergoes one epoch of pretraining, when applicable, followed by one epoch of finetuning. Across experiments, we retain LLaVA-1.5’s default hyperparameters, including learning rate and batch size, so that observed differences can be attributed to the ablated factors rather than to auxiliary optimization choices.

Fine-grained performance is reported as the average score across the four classification benchmarks described in Section 3.2. For general VQA performance, we report the average score over three widely used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These benchmarks were selected because they combine broad domain coverage with widespread adoption in VLM evaluation.

4.2 BASE MODELS

The analysis in Section 3 indicates that recent models typically incorporate newer base LLMs, although many still rely on the CLIP ViT-L/14 vision encoder from Radford et al. (2021). We therefore examine how these base-model choices specifically affect fine-grained classification performance.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as the baseline, we ablate the LLM component across multiple vision encoders and pretraining configurations. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023), which is used in LLaVA, against Qwen2-7B (Yang et al., 2024), which is used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024), as well as Llama-2-7B (Touvron et al., 2023), Vicuna’s base model, and Qwen2-7B-Instruct, the instruction-tuned variant of Qwen2-7B.

Findings. As shown in Figure 4, replacing Vicuna-7B with Qwen2-7B substantially improves performance across all configurations, yielding average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. The instruction-tuned Qwen2-

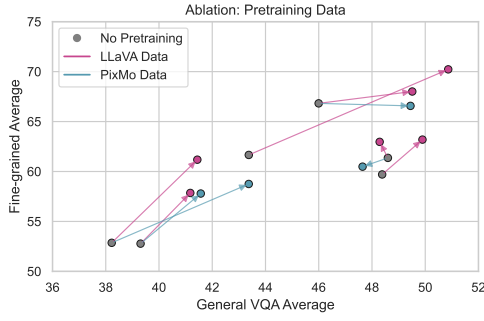


Figure 6: **Ablating pretraining data.** Introducing either LLaVA (CC-3M) or Molmo (PixMo-Cap) data produces similar gains: LLaVA data increases fine-grained and general scores by +4.0pp and +2.1pp, respectively, whereas PixMo data increases them by +2.4pp and +2.5pp.

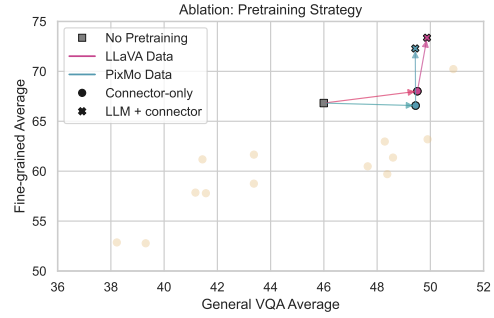


Figure 7: **Ablating training method by comparing connector-only pretraining with LLM+connector tuning.** Unfreezing the LLM during pretraining yields a +5.5pp improvement in fine-grained performance without adversely affecting general VQA scores.

7B-Instruct produces similar improvements (+7.5pp and +8.1pp, respectively). Llama2-7B, the non-instruction-tuned base model underlying Vicuna, yields more modest improvements, although the downstream implications of using a model without instruction tuning remain unclear. Importantly, all trained models, including the lower-performing Vicuna-based VLMs, consistently produce multiple-choice responses in the correct format. This pattern suggests that fine-grained classification differences arise from improved knowledge of the presented options rather than from superior adherence to the required response format.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as the vision encoder. By contrast, Qwen2-VL (Wang et al., 2024b) uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which demonstrates stronger zero-shot performance on object-recognition tasks such as ImageNet. We evaluate both vision encoders across several LLM choices and pretraining configurations.

Findings. Figure 5 shows that the effect of substituting DFN-CLIP H/14 for CLIP L/14 depends on whether connector pretraining precedes finetuning. Without pretraining, DFN-CLIP reduces general VQA performance (-2.5pp) while only modestly improving fine-grained benchmark scores (+2.8pp). For models that undergo connector pretraining, however, general VQA performance remains stable or improves slightly (+1.2pp), whereas fine-grained performance increases substantially (+4.5pp). This interaction indicates that stronger vision encoders can markedly improve fine-grained classification in VLMs, provided that they are adequately integrated during pretraining.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pre-training before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are commonly trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, uses large-scale image-caption datasets and a captioning objective to establish alignment between the vision encoder and the LLM. The second stage, instruction tuning, refines the VLM with carefully curated multi-round, multimodal instruction-completion

Table 1: **Summary of important ablation settings and comparison to existing models (bolded).** Upgrading the base LLM and vision encoder, together with pretraining the connector and LLM on sufficient data, improves fine-grained performance, whereas instruction finetuning has a smaller effect. This pattern contrasts with general VQA benchmarks, where the choice of LLM accounts for a larger share of the observed gains.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

data, aligning the model more closely with human intent in real-world applications. We examine how these two stages influence the fine-grained capabilities of VLMs.

4.3.1 PRETRAINING

Setup. We investigate several central questions about the pretraining process:

- Is pretraining necessary for LLaVA-architecture models when the base models have already been trained on billion-scale data?
- How does connector-only pretraining compare with unfreezing the LLM?
- How consequential is pretraining-data quality for downstream performance?

We consider three pretraining configurations: LLaVA pretraining data (Liu et al., 2023), a subset of CC-3M with web-scraped captions; Molmo pretraining data (Deitke et al., 2024), namely PixMo-Cap with high-quality, human-annotated detailed captions; and no pretraining. For most experiments, we follow Liu et al. (2023) and pretrain only the connector. We additionally examine a strategy similar to that of Deitke et al. (2024), in which the connector is pretrained for the first 20% of steps before the LLM is unfrozen for the remaining 80%. All configurations are subsequently subjected to one epoch of finetuning on LLaVA finetuning data.

Findings. Figure 6 summarizes the pretraining ablation results. Connector-only pretraining generally improves VLM performance, with the exception of the Qwen2 and CLIP ViT-L/14 combination. The gains are especially pronounced for Vicuna-based VLMs, which exhibit average improvements of +6.1pp in fine-grained classification and +3.1pp in general VQA performance.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Somewhat unexpectedly, pretraining on low-quality web-scraped captions (LLaVA) and on highly detailed human annotations (PixMo) yields only minimal differences. Relative to LLaVA data, PixMo data produces a -1.6pp change in fine-grained performance and a +0.4pp change in general VQA scores.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We hypothesize that connector-only pretraining may prevent the model from fully exploiting higher-quality captions. To test this hypothesis, we pretrain both the LLM and connector while incorporating a 20% connector-only warmup phase. Figure 7 shows that this strategy substantially enhances fine-grained classification performance (+5.5pp) without reducing general VQA scores. However, the improvement appears for both LLaVA and PixMo pretraining data, suggesting that training on lower-quality captions does not measurably impair the LLM’s language capabilities.

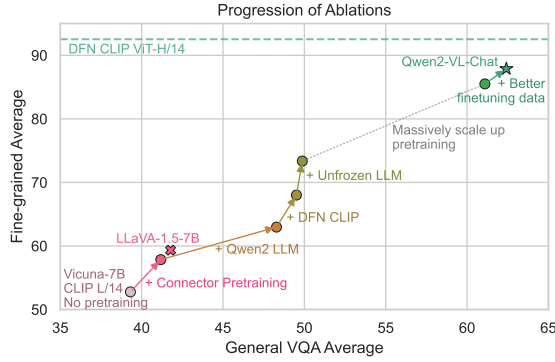


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B and CLIP ViT-L/14, successive modifications to the base LLM, vision encoder, and training settings raise fine-grained classification accuracy from 52.8% to 73.4%, accounting for more than half of the gap to Qwen2-VL-Chat’s 87.9%. We attribute the remaining difference primarily to Qwen2-VL’s extensive 1.4T-token pretraining corpus.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. Departing from the preceding setup, we begin with the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b), from which Qwen2-VL-7B-Chat is trained, and finetune it on LLaVA-Instruct data (Liu et al., 2023) according to the same experimental protocol. We compare these results with Qwen2-VL-7B-Chat, which uses 2M in-house examples for instruction finetuning (Wang et al., 2024b). This comparison helps isolate the contribution of different finetuning datasets.

Findings. The results indicate that finetuning solely on LLaVA finetuning data slightly reduces both fine-grained and general benchmark performance. Specifically, fine-grained scores decrease by 2.4pp, from 87.9% to 85.5%, while general benchmark scores decline by 1.3pp, from 62.4% to 61.1%. As illustrated in Figure 8, this is the smallest contribution to fine-grained performance among our ablations, suggesting that instruction finetuning is less central to fine-grained knowledge acquisition than base-model selection and pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize the ablation studies by showing how each component contributes to closing the performance gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Starting from the baseline LLaVA architecture (Liu et al., 2023), which combines Vicuna-7B (Chiang et al., 2023) with CLIP ViT-L/14 (Radford et al., 2021), we systematically modify components to increase fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The analysis shows that switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately enhance fine-grained capabilities relative to general VLM abilities, whereas replacing the LLM with Qwen2-7B (Yang et al., 2024) accounts for most of the improvement in general VQA performance.

Despite these substantial improvements, a notable 12-point gap in fine-grained classification remains unexplained by the ablations (Table 1). We consider two plausible sources of this discrepancy: architectural differences and pretraining-data scale.

Although architectural changes may contribute to performance differences, the observational results (Section 3.4) and training ablations (Section 4.3) suggest that pretraining-data scale is the dominant factor. Our experiments with LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) data use

comparatively small datasets, each containing fewer than 1M image-caption pairs, corresponding to approximately 200M and 400M tokens, respectively. By contrast, [Wang et al. \(2024b\)](#) report pretraining Qwen2-VL on 1.4T tokens, orders of magnitude beyond the scale considered in our experiments. This disparity in pretraining scale likely explains much of the remaining performance gap and underscores the importance of extensive pretraining for developing VLMs with stronger fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study has several practical limitations that also define natural directions for future research. Because of computational constraints, our training comparisons are limited to <1M data points rather than the billion (B)-scale regimes used by some newer VLMs, leaving unresolved how substantially larger-scale training might modulate the patterns observed here. In addition, newer work proposes alternative pretraining strategies that may interact differently with fine-grained visual understanding, and these interactions remain outside the scope of the present ablations.

In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks, showing that fine-grained visual classification is a crucial but underexplored dimension of VLM capability. Through detailed analysis of key model components and training paradigms, we identify strategies for improving fine-grained classification and strengthening vision-centric capability, thereby increasing the relevance of VLMs for real-world settings that demand precise visual understanding.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: Detailed results for the model ablation study.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: Detailed model-ablation results on general VQA benchmarks.

A FULL RESULTS

We report the complete ablation results for aggregated performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-H/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: Detailed model-ablation results on fine-grained classification benchmarks.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As described in Section 3.2, we evaluate VLMs and their base vision encoders on four traditional classification benchmarks:

ImageNet-1K (Deng et al., 2009): A foundational image-classification dataset, organized according to the WordNet hierarchy, that spans a broad set of supercategories, each containing multiple fine-grained subcategories. The ImageNet test set contains 50,000 images distributed across 1,000 classes.

However, the structure of the ImageNet/WordNet class hierarchy makes many classes too visually distinct from other classes to support hard multiple-choice construction; for example, only three classes appear under the “person” category. Because of this limitation, together with the large size of the dataset, we use human-annotated incorrect model predictions from Recht et al. (2019) to select images for which difficult negative answer choices can be constructed. Beginning with 19,056 annotated examples, we reduce the pool to 5,457 images spanning 928 of the original 1,000 classes. Because this curation leaves the remaining classes unevenly balanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A 102-species flower dataset characterized by substantial intra-class variation and inter-class similarity. The Flowers dataset contains 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A 37-category pet dataset that includes challenging variation in pose, lighting, and occlusion. The Pets dataset contains 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset comprising 101 food categories and exhibiting substantial within-class visual diversity. The Food dataset contains 25,250 test examples.

Question generation. For all fine-grained classification test sets except ImageNet, described above, we generate hard negatives by following Geigle et al. (2024) and performing zero-shot classification on the test examples with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022). We take the top 4 incorrect predictions and shuffle them with the correct label to produce the final answer choices.

Prompt formatting. In our multiple-choice VQA format, the question is always phrased as “What type of object is in this photo?” The answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because different VLMs are trained with different multiple-choice prompts and output formats, we use each model’s corresponding format from VLMEvalKit (Duan et al., 2024); for example, one format specifies: “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot evaluation procedure, so no question format is required. The class names passed to the text encoder are instead inserted into the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational analysis in Section 3, we rely directly on scores reported by the Open-VLM Leaderboard (Duan et al., 2024). The default leaderboard reports the average score across eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This reported average score is used only in Figure 2.

For all subsequent evaluations, including both existing models and models trained in Section 4, we restrict the comparison to a subset of multiple-choice VLM benchmarks to obtain a more direct performance comparison. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A 2,948-example dataset that tests a broad range of VLM capabilities spanning perception and reasoning. Although most examples concern coarse-grained perception and visual reasoning, some examples test “fine-grained perception”; however, their categories, such as action recognition, attribute recognition, and OCR, differ from the fine-grained classification domains considered in this work.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset covering a broad distribution of domains and applications. Although the test set contains 10,500 examples, most work reports model performance on the validation split, which contains 900 questions. MMMU evaluates general knowledge grounded in visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): A dataset comprising 1,500 questions selected by human annotators. The examples are chosen to cover a diverse set of tasks, both across domains and across low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). All existing models have predefined evaluation procedures; for our trained models, we apply the LLaVA evaluation code because all finetuning uses LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023). For most experiments, we retain the LLaVA architecture and training procedure to preserve comparability across ablations.

Pretraining. This procedure consists of one pretraining epoch on either the LLaVA pretraining data, obtained from CC-3M images and captions, or the Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we use only the “caption” field and exclude the raw audio transcript. During pretraining, the batch size is 256. In most pretraining runs, we tune only the randomly initialized MLP connector, using a learning rate of 1e-3. When ablating pretraining strategy, we first tune only the connector as a warmup for 20% of steps, after which we unfreeze and tune the LLM as well, using a learning rate of 2e-5 for the remaining 80% of steps.

Finetuning. For finetuning, we always train for one epoch on the LLaVA-1.5 finetuning data, which combines LLaVA-Instruct with VQA training data from several datasets to support proper benchmarking. During finetuning, the LLM learning rate is 2e-5 and the batch size is 128. The vision encoder remains frozen throughout both training phases.

For the ablation experiment on the Qwen2-VL architecture, we initialize from Qwen2-VL-7B Base and train with the same method, unfreezing only the LLM.

Compute. We train all models using 4 nodes, each equipped with 4 A100 (40GB) GPUs. Training typically completes within 20 hours, with runtime variation depending on the CLIP model size.

LLaVA vs. PixMo data. In the main text, we distinguish between the caption quality of LLaVA pretraining data and PixMo pretraining data. LLaVA pretraining data is sourced from CC-3M, which consists of web-scraped captions that are commonly characterized by low quality and limited relevance. PixMo-Cap, by contrast, is collected from human annotators who record a 60–90 second audio description of an image; these descriptions are then consolidated into image captions (Deitke et al., 2024). A surface-level analysis of the two datasets shows that LLaVA data averages 9.8 words per caption, whereas PixMo data averages 169 words per caption.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”